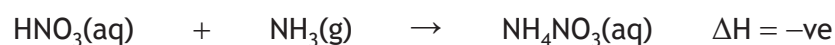


6. Ammonium nitrate is a commonly used fertiliser.

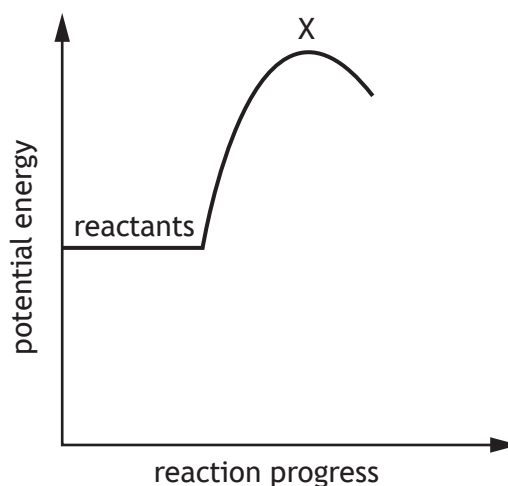
- (a) (i) (A) Ammonium nitrate is made industrially by adding nitric acid, HNO_3 , to ammonia, NH_3 .



Complete the diagram to show the shape of the enthalpy diagram for this reaction.

1

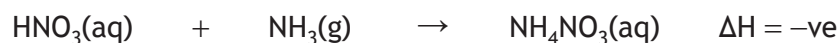
(An additional diagram, if required, can be found on page 40.)



- (B) State the term for the unstable arrangement of atoms formed at the point labelled X on the potential energy diagram above.

1

- (ii) 1316 litres of nitric acid of concentration 9.5 mol l^{-1} was reacted with 220 kg of ammonia ($GFM = 17 \text{ g}$) to produce ammonium nitrate.



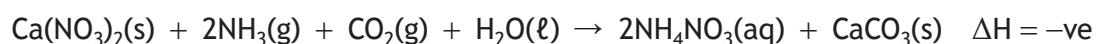
Show, by calculation, which reactant was in excess.

2



6. (a) (continued)

(iii) Ammonium nitrate can also be produced by the reaction

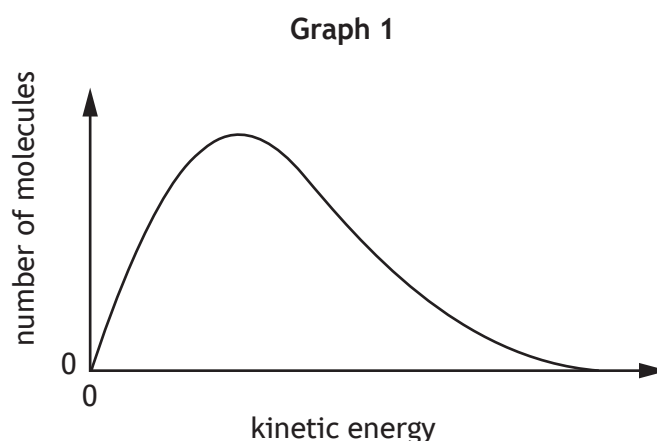


Suggest a reason why the method shown in part (a) (ii) is the preferred industrial route.

1

(b) The rate of reaction can be altered by changing the temperature or using a catalyst.

(i) **Graph 1** shows the distribution of kinetic energies of molecules in a gas mixture.



(A) Suggest what is represented by the area under the curve in **Graph 1**.

1

(B) Add a second curve to **Graph 1** to show the distribution of kinetic energies at a higher temperature.

1

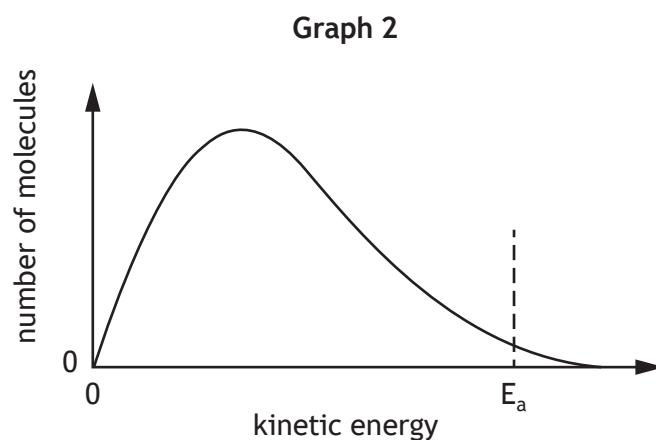
(An additional graph, if required, can be found on *page 41*.)

[Turn over



6. (b) (continued)

- (ii) In **Graph 2**, the activation energy for the reaction, E_a , is marked on the x -axis.



Draw a line on **Graph 2** to show how a catalyst affects the activation energy, E_a .

(An additional graph, if required, can be found on *page 41*.)

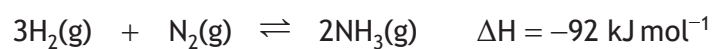
1



* X 8 1 3 7 6 0 1 2 2 *

6. (continued)

- (c) Ammonia used to produce ammonium nitrate can be made by the Haber process.



- (i) Reversible reactions, such as the Haber process, can reach a state of dynamic equilibrium in a closed system.

State what is meant by the term dynamic equilibrium.

1

- (ii) (A) The ammonia produced is continuously removed.

Explain how this will affect the production of ammonia.

2

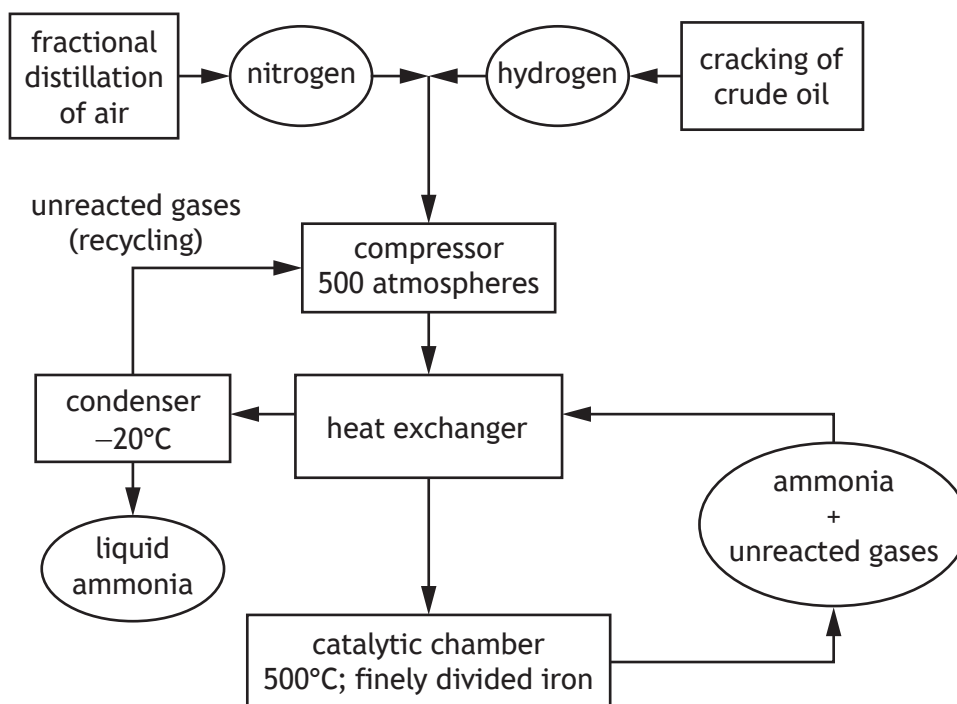
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6. (c) (ii) (continued)

(B) A flow diagram of the Haber process is shown.

One way to reduce costs in the process is to use a heat exchanger to transfer excess heat from one part of the process to use in another part of the process.



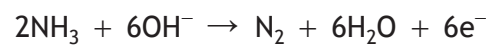
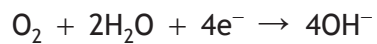
From the flow diagram, state another way that the manufacturing process maximises profit or minimises the impact on the environment.

1

6. (continued)

(d) Ammonia is currently being investigated for use in fuel cells.

The reactions taking place at the electrodes are



(i) Write the overall redox equation for the reaction taking place in the fuel cell.

1

(ii) Identify the reducing agent in the reaction taking place in the fuel cell.

1

[Turn over

